

5.1 Generate the coefficients of an FIR High-Pass filter using Rectangular windowing techniques.

```
clc; clear; close all;

%%% --- Input Parameters ---

fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)

%%% --- 1. FIR High Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'high', rectwin(N+1));
disp('--- High-Pass Coefficients (Rectangular Window) ---');
disp(b_rect);

%%% --- 2. FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);

%%% --- 3. FIR High Pass using Hanning Window ---
b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann);

%%% --- Frequency Response Plots ---

figure;
freqz(b_rect,1);
title('High-pass FIR using Rectangular Window');

figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');

figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');
```

